

Cost Is Only Half the Decision: Users Cannot Forecast Small-Model Success

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Abstract

Interfaces that only show the energy cost of an AI request are missing half of what a routing decision needs: whether the routed model could actually answer the request. Converting MMLU-Pro into 8,454 open-ended questions that more closely match real-world queries, we estimate Gemma-4-E2B-it’s reference-answer success by repeated generation and panel grading. This yields the absolute success probability of the specified local model as a target to compare routing strategies. In a preregistered study, 30 participants supplied 840 forecasts across 280 held-out questions, and we compare these forecasts with predictions from a ModernBERT classifier on those same questions. Aggregated human forecasts overestimated local success by +0.95 of 6 bands and participants disagreed with one another (Krippendorff’s $\alpha = 0.166$). These results suggest unaided judgment is unreliable in this task. The classifier agreed better with the target ($\kappa_w = 0.237$ against 0.049). On 1,375 test questions, no threshold met both error limits on false local selections and unnecessary cloud selections. In an electricity scenario with an ideal perfect cloud reference, classifier routing on the $n = 280$ held-out queries preserves more answer success (80.4%) than human routing (59.3%) at the cost of additional watt-hours per query (48.85–129.29 vs 20.50–54.47), yet at a fraction of no routing at all (101.48–269.84). This direction holds on the $n = 1,375$ test set (84.7% answer success; 59.14–158.38 vs 108.17–288.74). Estimates for model consumptions come from the EcoLogits library, and electricity savings depend on generation-cost reference and retry behavior. Together, these findings motivate evaluating whether automated success forecasts can support model choices on everyday requests.

1 Introduction

Generative AI is increasingly used in everyday work. By May 2026, 45% of US workers aged 18–64 reported using it for their jobs [6, p. 12]. An analysis of consumer ChatGPT use found that writing assistance, information seeking and practical guidance together account for about 77% of messages [11]. This adoption has increasingly brought AI’s energy demands into focus. While both training and inference consume electricity, inference now dominates AI-related energy use [29, p. 31]. But serving a request incurs an electricity cost that varies with the model and workload [35]. The benefits of chain-of-thought prompting depend on the task [45]. Long-reasoning models can repeat solutions on simpler mathematical problems, adding computation with little improvement in accuracy [13]. Measurements across eight open models on the same dedicated machine show that energy use varies with both the model and task category [27].

Letting users choose a lower-cost model presents two interlocking challenges: First, informed choice requires transparency about which model processes a request, how resource-intensive that processing is, and whether a lighter execution path is available. Moreover, the objectives that provider-side optimisations target need not coincide with end-user preferences. A provider optimising for aggregate cost, latency, or throughput may make trade-offs that a sustainability- or quality-conscious user would not choose. Theoretical work on routing shows how provider cost and user preferences over success and delay can favour different routes [37]. A device-edge router makes a related trade-off explicit by conditioning routing decisions on an accuracy–cost weight [51]. The second challenge is an intention–action gap. When running a model locally requires a separate application and manual model selection, users must leave their established workflow to choose a local alternative. These challenges motivate an interface that makes resource costs visible and local alternatives accessible within the same workflow.

Some systems let users define routing policies [47], while others expose route metadata to developers [40]. Browser extensions and chat prototypes surface estimates of AI energy use and associated emissions [9, 25, 42], including alongside choices presented as differing in energy use and performance [31, 41]. For someone deciding whether to run a request locally, cost is only half the choice: they also need to know whether the local model is likely to answer the request well enough. We study that judgment in the setting of *LocalGate*, a proposed design in which requests are routed to a locally hosted model from inside a commercial chat interface. The decision must be made from the prompt alone, before any answer exists, and a failed local attempt followed by a cloud retry incurs both costs. This paper examines how reliably people and a classifier can make that judgment and what acting on it would mean for electricity use.

Our work is guided by the following research questions:

- RQ1: How well can people judge whether a prompt can be answered acceptably by a small, locally run language model?
- RQ2: How does classifier performance compare to human judgment on the same prompts?
- RQ3: How would classifier-based routing decisions affect electricity use and answer quality compared with all-cloud inference?

Predicting success from the prompt alone can be difficult. In Fang et al.’s device–cloud experiments, a prompt-only router baseline fit its training data closely but generalised poorly on mathematical questions; the authors attribute this gap to surface features being an imperfect guide to problem difficulty [20]. Comparing people and a classifier on the same prompts requires a common target, which we obtain by constructing open-ended questions from MMLU-Pro

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and estimating the local model’s success on each through repeated generation and grading. Both forecasts then predict that same quantity. We finally model how routing on such forecasts would change electricity use and answer success under stated cost and retry assumptions, which the following sections outline.

2 Related Work

2.1 Inference Electricity and Query Routing

The electricity needed for inference varies across models and tasks [35]. Understanding that cost also means looking beyond the model itself. Elsworth et al. use Gemini Apps telemetry to examine production inference, including datacentre overhead but excluding training and end-user devices [19]. Where direct measurements are unavailable, EcoLogits estimates impacts from output-token counts and assumptions about the model and serving infrastructure [17, 43].

Query routing aims to use a less costly model when its answer is likely to be adequate. Some approaches account directly for energy use. GreenServ learns routing policies from accuracy and measured GPU inference energy [53]. Hoffmann and Majuntke use measurements from similar benchmark prompts to choose among eight open models. On 25 held-out prompts, their router achieved 72% accuracy with 44% of the reported software carbon intensity of Llama 3, which achieved 88% accuracy [27]. Other approaches consider API spending, as in FrugalGPT [12], or the fraction of queries sent to the smaller model, as in Hybrid LLM [15]. These measures describe financial cost or model use, rather than electricity consumption.

Routing can take place before generation or after an initial answer has been assessed. FrugalGPT scores an answer using the query and the generated response, then decides whether to accept it or continue to another model [12]. RouteLLM instead predicts from the query whether the stronger model will win a preference comparison against the weaker one [39]. Hybrid LLM predicts the probability that the smaller model meets a quality criterion relative to the larger one. Its probabilistic variant constructs training labels from repeated generations to account for variation in answer quality [15]. For device-edge routing, CR² uses a teacher’s estimates of each model’s correctness together with execution costs to train a gate that chooses between local and edge execution [51].

In this study, human participants and the classifier both predict the chance that a specified local model will produce an answer matching the reference. Their forecasts can therefore be compared against the same outcome. The electricity analysis then examines the consequences of using those forecasts to route requests. Table 7 in Appendix D.6 compares the timing and prediction targets of the routing approaches.

2.2 Eco-Feedback, Transparency and Control

Eco-feedback helps people see the resource consequences of their actions. In a field experiment on showering, Tiefenbeck et al. found that displaying consumption during use reduced resource consumption [46]. For AI, much of the available feedback has instead been designed for technical users. A 2025 review of 16 machine-learning eco-feedback tools found that they mainly served developers and researchers, with a strong emphasis on model training [24]. End-user interfaces bring this information into decisions about everyday

AI use. How these choices are presented also matters. Preselected options are chosen more often on average, although the effect varies substantially across studies [30]. For LocalGate, the default route and how users can override it are therefore an important part of its design.

Some interfaces present resource information alongside a choice of model. In an online experiment, Klesel and Messer found that an EU-style energy label increased selection of an option advertised as efficient but lower-performing from 4.7% to 39.3%. In reality, both options used the same underlying model [31]. Patkar et al. studied an interface with an energy-mode switch over five days with 11 users. Using the recorded token profiles, they estimated that the observed mode choices required about 47% less energy than assigning every request to Performance mode [41].

Feedback may also influence how much people use AI. In the week-long GPTFootprint study, all nine student participants reported greater awareness, but the total number of queries did not show a decline [25]. EcoChat followed two baseline weeks with two weeks of feedback. Its estimate suggested fewer weekly prompts, although the uncertainty interval also included a small increase. Interviews showed that participants reflected on appropriate use while work demands sometimes limited their ability to change it [42]. In a university study, participants with a tracker used fewer input tokens per prompt and had lower estimated emissions per prompt. However, the tracker and comparison groups did not differ significantly in the relative changes in each participant’s input-token use and estimated total emissions between the first and last eight days [2]. Greater awareness, choosing a different model and reducing total use are thus distinct outcomes of eco-feedback.

Resource information alone does not tell users whether a model will answer their request successfully. Providing confidence information may help them decide when to rely on it, but changes in reliance do not necessarily improve decisions. In an income-prediction experiment, displaying confidence made participants’ reliance track the AI’s confidence, without significantly improving human-AI accuracy [52]. Interfaces can also encourage users to reconsider advice. In a nutrition task with simulated AI advice, designs intended to encourage deliberation reduced acceptance of incorrect advice about which ingredient was the carbohydrate source. Participants, however, rated these designs as more complex than interfaces offering simple explanations [8].

These studies examine how people respond to resource information, predictions or advice supplied by an AI system. The interfaces studied by Klesel and Patkar provide energy and general performance cues, but as outlined previously, they do not estimate whether a model will answer a particular request successfully. Vafa et al. studied people’s predictions of LLM correctness, how these changed after learning whether another question was answered correctly, and how well these expectations aligned with model performance [48]. Our study compares human forecasts with a classifier trained to predict a specified local model’s success, evaluating both against the same reference estimates from repeated generation.

2.3 From Multiple-Choice to Open-Ended Evaluation

Many benchmarks, including MMLU-Pro, are multiple-choice, but selecting an answer from a list differs from producing one without supplied choices [5]. Models can select correct options above chance even without seeing the question [3, 4]. Removing the options therefore helps examine whether a model can produce a correct answer without these clues.

But evaluating such answers requires more than changing the answer format. Some questions rely on information contained in the options and no longer make sense without them. Myrzakhan et al. screen for questions that remain answerable without options and use the original correct answer as a reference for grading generated responses [38]. However, a question may remain answerable yet admit several distinct correct answers, which one original option cannot cover. Myrzakhan et al. encountered this problem with HellaSwag, where several plausible continuations could be valid, and excluded it from their overall leaderboard score.

For questions with an unambiguous answer, grading must still recognise that the same meaning can be expressed in different words. Chandak et al. evaluate answer matchers, models that compare a generated response with a reference answer in the context of the question. On MMLU-Pro and GPQA-Diamond subsets screened for unambiguous answers, recent matchers agreed more closely with human grading than either multiple-choice evaluation or judges given no reference answer [10]. Our conversion checks address this answerability and notion of preserved meaning, while the grading step tests whether responses convey the meaning of the original correct option. Both participants and the classifier therefore predict the same outcome: whether the local model can produce an answer that matches the reference in meaning, without seeing the answer options. A common target.

3 Data & Methods

3.1 Data

Routing requires a way to establish what a local model can handle. A question’s category alone does not reveal whether that model can answer it, and a question that is easy for one model may be difficult for another. We therefore define an item’s difficulty through the model’s performance on it: the proportion of satisfactory answers across k generations is its estimated success rate, $\hat{p}$. A *satisfactory* answer states the reference answer without contradiction or omission, irrespective of wording, length or style (Appendix B.2). The procedure is repeatable across models, but logically each label remains specific to its model and generation-and-grading protocol.

Corpus. MMLU-Pro [49] provides 12,032 exam-type multiple-choice questions across 14 categories and was developed partly to correct errors in the original MMLU. The text of each question’s correct option serves as the reference for grading its open-ended responses. Conversion must therefore preserve what the question asks, so that the original answer remains appropriate after the options are removed.

Conversion. The deepseek-v4-flash [14] model was used to apply two filters and a rewrite. In a preliminary comparison, it made fewer false acceptances than Mistral Small and changed fewer

filtering decisions between repeated runs. The first filter dropped option-dependent items, such as “none of the above”. The second filter reconsidered rejections and readmitted questions answerable with the intended answer alone. Rewriting the survivors to stand alone left 8,454 questions. The three authors completed a three-part human validation of the conversion and automated grading, using dedicated annotation interfaces hosted on Hugging Face Spaces. First, they assessed whether 248 original questions could be answered without options, without seeing the automated filter decisions or rewrites. Second, they reviewed 152 retained rewrites to check that each preserved the original meaning and could be understood without the answer options (Appendix B.5).

Generation and grading. Each converted question received $k = 5$ sampled answers from the standard Gemma-4-E2B-it checkpoint [23], served under vLLM [32] with $T = 1.0$, top- p 0.95, top- k 64 and zero-shot prompting. Repeated sampling captures variation in success. Three open-weight LLM judges from different families graded each answer by 2-of-3 majority. None evaluated its own output or answers to questions written by its family. Judges checked whether responses stated the reference answer and counted only an explicit *match* as correct, rather than solving questions themselves. The full rubric appears in Appendix B.2, with prompts, schemas and data linked in the repository. In the third part of the human validation, the authors graded 100 model responses to check agreement with the automated panel. The panel agreed with their majority on 90 (Scott’s $\pi = 0.79992$). Bulk grading proceeded despite narrowly missing the 0.80 calibration threshold (Appendix B.4 documents the sampling change and decision).

Scale and Splits. For comparison with human forecasts, $\hat{p}$ is mapped to six equal-width bands, $\ell = \min(\lfloor 6\hat{p} \rfloor, 5)$. This puts the reference labels, human ratings and classifier predictions on the same scale, with band 0 covering success probabilities below 1/6. Agreement is measured with linearly weighted κ_w , which accounts for chance agreement and penalises disagreements in proportion to the number of bands between them. For binary routing, a question is classed as locally solvable when at least half its sampled answers are satisfactory ($\hat{p} \geq 1/2$). Routing accuracy measures agreement with this local/cloud label, not how often a new answer would be correct. The converted pool was divided into 6,705 development questions and 1,655 test questions, with 94 excluded to prevent overlap (Appendix D.2). From the test set, 280 questions were selected for the human study, 20 per category, balanced across the initial success-rate bands. The other 1,375 questions provide a broader test of the classifier and its routing thresholds (See Appendix Table 3 for set and role summary). The five-generation estimates served two purposes, training the classifier and selecting study questions across a range of difficulties. For the paired human–classifier comparison, 30 fresh answers per study question provided a more precise reference estimate. This exceeded the preregistered 20 (Appendix B.3). Both forecasters were evaluated against these same new labels.

3.2 RQ1: Human Routing Judgment

The study involved 30 participants. Among the 29 who supplied age and gender, mean age was 23.3 years (SD 2.9, range 18–31), with 20 male and 9 female participants. Missing or ambiguous background

responses did not exclude anyone’s forecasts. Table 4 reports the full demographic summary with a valid-response denominator for each field.

Before rating, the instructions named Gemma-4-E2B-it as the model to assess and explained local versus cloud execution, describing local processing as private and less energy-intensive. Participants then forecast how likely it was to answer each question satisfactorily, without solving the questions, seeing model answers or receiving accuracy feedback. Instructions encouraged instinctive judgment. Each participant rated 28 questions, two per category, giving three forecasts per question. The six probability bands matched the reference scale and offered neither a neutral midpoint nor abstention. Participants did not see reference answers, success rates, category labels or the proportion locally solvable (Appendix C.1).

Hypotheses and evaluation. The human study was preregistered on OSF¹ with one confirmatory hypothesis about agreement and two opposing exploratory hypotheses about the direction of routing errors:

- H1: Users’ judgments of local solvability diverge from the model’s actual performance; agreement with ground truth is at best “slight” (following Landis and Koch [33], $\kappa_w \leq 0.20$). This is tested against a preregistered adequacy floor of $\kappa_{\min} = 0.50$.
- H2: The disagreement is directional, with users routing more prompts to the cloud (exploratory).
- H3: The disagreement is directional, with users routing more prompts to the local model (exploratory).

Forecasts were analysed by question ($n = 280$), taking the median of the three ratings as the human forecast for each question. The median minimises absolute error, matching the linear weighting of κ_w , so we use it instead of a rounded mean. To derive a local/cloud routing decision, we classified each rating at the same probability boundary of $1/2$ and took the majority vote.

For H1, we test $H_0: \kappa_w \geq \kappa_{\min}$ at one-sided $\alpha = 0.05$. To estimate uncertainty, we resampled the 280 questions with replacement 10,000 times and calculated bias-corrected and accelerated bootstrap intervals [18]. The cohort selection and bootstrap procedure were chosen during analysis (Appendix B.3); the resampling keeps the participant group fixed.

For H2 and H3, one-sided paired McNemar tests compare those error counts at $\alpha = 0.025$ each. Further analyses examined inter-rater agreement, rater-exclusion sensitivity and exploratory background associations, which had limited power with 30 participants (Appendices C.3 and C.5).

3.3 RQ2: Automated Routing Classification

The classifier learns to predict local-model success, using five-generation success counts as training targets. Its only input is converted question text, without options, reference answers or category labels.

ModernBERT-base [50] encodes every converted question in full, with attention projections fine-tuned by low-rank updates (LoRA) [28]. The encoder feeds an output head that turns its representation of the question into a success forecast. We compared

seven such heads, including binomial and beta-binomial models of the number of successful answers. Cross-validation kept related questions in the same fold, preventing text families from appearing on both sides of a training-validation split.

We selected the model with the lowest mean per-generation log loss, which rewards accurate probability forecasts and penalises confident errors. This matters because routing needs useful success probabilities, whereas selecting on band agreement (κ_w) alone could favour extreme predictions at the expense of calibration. The selected beta-binomial head was then refit on all 6,705 development questions. Its output is a beta distribution whose mean gives the predicted success probability (Appendix D.3).

Evaluation. One evaluation uses the 1,375 test questions outside the human study. Our primary metric is linearly weighted κ_w , measuring agreement with reference success-rate bands. Further measures assess whether predicted probabilities match observed success rates (calibration) and count routing errors when a predicted success probability of at least $1/2$ sends a question locally. An always-cloud baseline shows what routing accuracy is achievable without distinguishing questions. For band agreement, it assigns every question to the lowest band, giving $\kappa_w = 0$ because a constant forecast adds no agreement beyond chance. A second baseline assigns each question its category’s mean success rate across the 6,705 development questions, testing what category alone can tell us.

Beyond average forecast accuracy, we examined whether a single routing threshold could satisfy a chosen operating target. The target allowed at most 10% of questions classed as not locally solvable to be sent locally (false positives), and at most 20% of locally solvable questions to be sent to the cloud (false negatives).

The paired evaluation compares classifier and human forecasts on the 280 study questions, using the same $k = 30$ reference labels for both. Uncertainty in their κ_w difference is estimated by resampling questions together with both forecasts (paired BCa bootstrap, 10,000 resamples, two-sided $\alpha = 0.05$). The classifier’s routing threshold stays at $1/2$, matching the reference definition without tuning it on these questions. Appendix D.4 gives the remaining evaluation details.

Because the study uses open-ended questions, a supporting comparison examined whether a router trained on multiple-choice questions could also forecast success in this setting. This router and the selected open-trained classifier were evaluated on all 1,655 test questions in both formats.

3.4 RQ3: Electricity Scenarios

RQ3 compares human and classifier routing on the 280 study questions, then classifier routing on the 1,375 remaining questions (for which human performance is not available), overall and by category. The classifier threshold is $1/2$. Each question retains its observed success rate ($k = 30$ or $k = 5$, respectively) and mean answer length.

We compare these policies with an ideal perfect cloud reference, assigning it GPT-5.5 Pro (a frontier model with available estimates) electricity costs. Perfect success is an assumption, not observed cloud performance. The main setting accepts the selected model’s first answer, so local failures reduce expected success. A recovery

¹https://osf.io/zx5ej/overview?view_only=71c641fc77a5464aa207c976dfc2f492

setting instead permits zero to five extra local attempts upon failure, then falls back to the cloud.

EcoLogits 0.11.1 supplies generation costs [16, 43]. gemma-4-26b-a4b-it supplies a local electricity proxy because E2B is not covered, while success rates remain those observed for E2B. Cloud output length initially matches local length. Electricity ranges reflect the lower and upper EcoLogits estimates. Classifier inference adds a measured 0.000221 Wh per request above idle. The comparison excludes human judgment and various costs (such as overhead service for cloud output, shared interface, and hardware production costs).

We check exact expectations against 10,000 Monte Carlo repetitions, assuming independent attempts at each question’s success probability. Appendix E gives the equations, overhead benchmark, category results and sensitivity to output length and cloud cost.

4 Results

4.1 Sample and Data Quality

The filter keeps roughly seven items in ten. Technical subjects survive more often than law or history, and the retained questions are easier for Gemma-4-E2B-it than those excluded. On the same 8,454 retained questions, the mean success count was 3.28/5 in multiple-choice form and 2.36/5 in open-ended form. This descriptive contrast combines differences in answer options, conversion and evaluation format (Appendix D.1 gives the full distributions).

All 30 participants contributed complete forecast sets, giving 840 forecasts across the 280 questions. Participants received their assigned questions in the specified order and all passed the attention check. The median time was 12.2 seconds per question and 8.1 minutes for the full rating task. The reference labels classed 48.9% of questions as locally solvable using the registered probability boundary of 1/2 (Appendix C.2).

Fresh generation changed the estimated difficulty of some study questions. The resulting sample no longer had the balance of success-rate bands used for selection, although most binary local/cloud labels stayed the same (Appendix C.4).

4.2 RQ1: Human Forecasts

The median human forecasts agreed only slightly with the reference labels ($\kappa_w = 0.049$, 95% BCa CI $[-0.006, 0.103]$). The bootstrap analysis gives a one-sided 95% upper bound of 0.094, well below the preregistered adequacy threshold of $\kappa_{\min} = 0.50$ and consistent with H1. These forecasts overestimated success by +0.95 bands on average, with a mean absolute error of 1.92 bands on the six-band scale. The reference places 82 questions in the lowest band, yet none of the median human forecasts falls there. That band was selected in only 16 of the 840 individual judgements. Questions in the lowest and highest reference bands received similar distributions of human forecasts, despite their very different observed success rates (Figure 1, Appendix C.3). Participants also disagreed with one another (ordinal Krippendorff’s $\alpha = 0.166$), so the aggregate overestimation coexists with substantial variation between raters.

Direction of routing errors (H2/H3). Converting the forecasts to routing decisions would send 80.0% of questions locally, compared

Table 1: Human and classifier forecasts on the same 280 questions, $k = 30$. Human agreement uses the median of three forecasts, while routing uses their majority decision.

	κ_w	Routing accuracy	Sent local
Human aggregate	0.049	51.8%	80.0%
Classifier	0.237	62.5%	48.6%
Always cloud	0.000	51.1%	0%
Locally solvable	—	—	48.9%

with 48.9% under the reference labels. The excess of 31.1 percentage points supports H3 rather than H2 ($p < 10^{-14}$, one-sided McNemar). The direction reverses the preregistration’s expectation that frequent frontier-model users might *under*-rate a small local model.

Removing each rater in turn, or excluding raters with unusual agreement scores, left agreement low and the excess of local selections intact (Appendix C.4). Background analyses found no reliable predictor of forecast quality, but the small sample leaves these associations imprecisely estimated (Appendix C.5).

4.3 RQ2: Classifier Performance

Comparison with human forecasts. On the same 280 prompts, against the $k = 30$ reference labels, the classifier reaches $\kappa_w = 0.237$ against 0.049 for the human aggregate, a paired difference of +0.188 (95% BCa interval $[+0.108, +0.271]$, excluding zero).

The classifier also has higher routing accuracy than both the human forecasts and the always-cloud baseline (Table 1).

Performance on the remaining test questions. On the 1,375-question test remainder, using $k = 5$ labels, the classifier reaches $\kappa_w = 0.378$, against 0.179 for the category mean and 0 for always-cloud. Routing accuracy is 72.0% against 66.5% and 52.4% respectively. Predicted probabilities broadly tracked observed success rates, with some overconfidence around 0.65. Expected calibration error was 0.033 using ten equal-width probability bins (Appendix D.4). At the predicted success cutoff of 1/2, the classifier exceeds both error limits on the 1,375 remaining test questions. It sends 27.7% of questions labelled not locally solvable to the local model (a 10% ceiling), while sending 28.3% of locally solvable questions to the cloud (a 20% ceiling). Raising the cutoff to 0.70 reduces false local selections to 9.7%, but selects only 28% of questions locally. Among those selected, 81.8% meet the reference definition of locally solvable, meaning that at least half their sampled answers succeeded. This proportion describes the selected questions, not an 81.8% answer-success rate. No threshold meets both error limits, and the minimum sum of the two error rates is about 54% (Figure 3, Appendix D.4).

4.4 RQ3: Conditional Electricity–Quality Trade-offs

On the 280 study questions, human routing uses less electricity (in Wh/question: 20.50–54.47 Human vs 48.85–129.29 Classifier; both a fraction of defaulting to cloud 101.48–269.84) because it sends many more questions locally, but accepts substantially more

Table 2: Local success follows the observed generation rates, while cloud answers are assumed correct. Electricity ranges use GPT-5.5 Pro costs from EcoLogits with equal local and cloud output lengths (Section 3.4).

Policy	Success (%)	Electricity (Wh/question)	Saved (%)
<i>280 study questions</i>			
Always cloud	100.0	101.48–269.84	0.0
Human aggregate	59.3	20.50–54.47	79.8
Classifier	80.4	48.85–129.29	51.9–52.1
<i>1,375 test remainder</i>			
Always cloud	100.0	108.17–288.74	0.0
Classifier	84.7	59.14–158.38	45.1–45.3

failures (Table 2): classifier routing preserves 21.1 percentage points more expected answer success than human routing.

On the 1,375 remaining questions, classifier routing retains 84.7% expected success at 59.14–158.38 Wh per question, against 108.17–288.74 for all-cloud inference, saving 45.1–45.3%. Recognising each local failure and sending it to the cloud (where success then reaches 100% by assumption) instead raises electricity use to 77.23–206.92 Wh per question. Category-level results and additional retries appear in Appendix E.

Absolute savings depend strongly on the cloud cost reference. With standard GPT-5.5 costs, the same first-answer policy on the test remainder uses 1.93–4.00 Wh instead of 3.48–7.24 Wh for all-cloud inference, saving 44.5–44.8%. Across both cloud references, all three output-length settings and the local retry limits, classifier routing saves electricity overall under the perfect-cloud assumption.

5 Discussion

6 Discussion

Participants struggled to anticipate E2B’s capabilities. They overestimated how often the model would succeed and disagreed about its chances of answering individual questions. The eco-feedback studies reviewed above show how cost information can raise awareness or influence model choice [31, 42]. Our results highlight another difficulty in making that choice: judging whether the model will answer successfully. Whether greater experience with AI tools or feedback could improve these forecasts remains untested.

The classifier predicted whether E2B would produce an answer matching the reference more accurately than the aggregated human forecasts. This complements routing research that predicts relative model quality [15, 39]. More accurate forecasts, however, do not necessarily mean lower electricity use. In the paired comparison, routing based on human forecasts used the local model more often, saving more electricity under the perfect-cloud assumption but accepting lower expected answer success. Classifier routing used the cloud more often and retained higher expected success. Choosing between these policies therefore depends on both the electricity budget and the level of answer success required.

7 Limitations

The findings concern Gemma-4-E2B-it under the reported settings and a small, predominantly student sample. The questions are exams rather than everyday requests, and reference-answer matching does not measure user satisfaction. Balancing the 280 study questions across categories and difficulty also means that routing rates need not resemble everyday use. The analysis concerns individual, independent LLM requests and may not generalise to multi-step agentic tasks. Moreover, the instructions described local execution as private and less energy-intensive, with no neutral condition for comparison, and supplied a shorter answer criterion than the full judges’ rubric. The framing may have influenced forecasts, but the study cannot measure its effect. Participants predicted an agreed routing based on success; they did not themselves choose the routing strategy.

Reference labels and analyses depend on answer keys, conversion and grading. Errors or ambiguity in the keys can affect these labels, as audits of the original MMLU illustrate [21]. The panel narrowly missed its calibration threshold. Its non-random sample repeated some questions and identical answers, limiting generalisability (Appendix B.4). Agreement on conversion self-containedness remained low after correcting reversed response labels (Appendix B.5). The changed bands between the $k = 5$ and $k = 30$ runs also show uncertainty in these reference estimates. Benchmark exposure during pretraining may also affect local success and the classifier’s forecasts (Appendix D.1). Other models and deployment settings require fresh validation.

Finally, the electricity scenarios assume that the cloud model always answers correctly and, in the recovery scenarios, that every wrong local answer is identified and retried locally or sent to the cloud. They rely on modelled generation costs, output lengths and retry behaviour. We expect the direction of the result to hold, and its magnitude to be understated by construction, for two reasons. First, the proxy for the local model is more energy-intensive than the model it stands in for, while the local model’s performance is maintained. Second, the cloud model is treated as an ideal oracle, and its cost is bounded by a capable current model for which EcoLogits provides an estimate [43]. EcoLogits itself rests on uncertain infrastructure assumptions, which further disadvantages our proxy relative to real use on local hardware. Neither expectation has yet been tested against real user behaviour. These scenarios therefore describe conditional operational trade-offs and do not demonstrate savings in a deployed system.

8 Conclusion

Cost information alone cannot tell users whether a local model will answer a request adequately. In this study, participants overestimated the model’s success and disagreed about individual questions. The classifier forecast success better than their aggregate, but no threshold met both chosen error limits on the broader test. In the illustrative scenario, classifier routing preserves more answer quality than human routing while using less electricity than all-cloud inference. The trade-off depends on generation costs and retries. Whether these forecasts improve everyday model choices remains to be tested.

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A Supplementary Discussion: Open Questions for LocalGate

LocalGate provides a basis for exploring several open questions about how people choose between local and cloud models. One direction is classifier-led routing that displays the estimated chance of local success and lets users override the decision. An override could express context unavailable to a prompt-only classifier, such as urgency, confidentiality or definite high quality answers without relying on users to forecast success more accurately. Adjustable thresholds could also let users express how much answer quality they are willing to trade for lower electricity use. Whether these features help users make better routing decisions remains an open question. As the study discussed in Section 2.2 illustrates, showing confidence can change when people rely on an AI system without necessarily improving their combined accuracy. LocalGate therefore needs to be evaluated on the routing decisions users make with its forecasts, as well as on the accuracy of the forecasts themselves.

The present study treats each question as a standalone request, whereas LocalGate would also encounter conversations. Evaluating first-turn routing requires excluding later turns from the input, since they would not yet be available when the decision is made. Routing later in a conversation could use earlier turns, either directly or through a summary, depending on the model and its effective context window. The summary would need to preserve relevant context, and both approaches would need testing to establish whether that context improves forecasts.

How users respond to automatic routing, overrides and resource feedback also needs to be studied over time. Household energy interventions show that responses to repeated feedback can change while conservation effects persist [1]. For LocalGate, this raises the question of whether users continue to notice its feedback and act

on it. Easier or cheaper local access might also encourage more requests, potentially offsetting savings per request through increased use [7]. Tracking both routing choices and total use would help establish whether LocalGate reduces overall electricity consumption effectively.

B Evaluation Protocol

B.1 Study Overview

Table 3 summarises the question sets used to develop the classifier and evaluate its predictions. The classifier was fitted and selected using 6,705 development questions. Of the 1,655 test questions, 280 were selected for the human study. On these questions, human and classifier forecasts were compared against the same reference success rates, estimated from 30 fresh model answers per question. The remaining 1,375 questions were used to evaluate the classifier and electricity scenarios without human forecasts.

In the table, k denotes the number of model answers generated per question, and k_h the number of human forecasts per question. Appendix D.2 explains how the datasets were split and how overlapping questions were excluded.

The conversion prompts and schemas are available with the converted dataset.

B.2 Judge Setup

The three AWS Bedrock judges were `ministral-3-14b-instruct` (Mistral), `gpt-oss-120b` (OpenAI) and `qwen3-next-80b-a3b` (Qwen). Google was excluded to avoid grading its own model’s output, and DeepSeek to avoid grading questions written by the converter. Each judge received the complete question, reference answer and model response, and returned a short explanation and verdict using a fixed prompt and schema. All three used temperature 0 and a limit of 400 output tokens per grading request, with `gpt-oss-120b` using low reasoning effort. Failed requests were retried, and every response received a valid verdict from all three judges before the panel decisions were calculated.

Judges received the following rubric.

You judge whether a response to an exam question states the same answer as a reference answer.

You are given a question, the reference answer, and a model’s response. The response may be long, may reason step by step, and may be cut off.

Apply one test. Find the response’s final answer — the answer it commits to, not answers it merely considers along the way — and decide whether that final answer states the same fact as the reference.

The reference answer is authoritative. Do not re-solve the question, and do not judge the response’s reasoning; if the response’s final answer disagrees with the reference, the verdict is `no_match` even if you believe the response. A response matches when its final answer contains everything the reference states. Wording never matters: paraphrase, a different language, different formatting, or extra correct detail alongside the reference content are all still a match (“Labrador” matches reference “dog”). For numeric answers, the response must agree with the reference to within 1% relative error, in any unit convertible from the reference’s; a range does not match a specific value, even if the range contains it.

A response is `no_match` when it commits to a final answer that contradicts or omits part of the reference, or when it hedges between several different final answers.

A response is `no_answer` when it never commits to a final answer: it is cut off before concluding, answers a different question than the one asked, is empty, or only restates the question. Nothing it says contradicts the reference; it simply never answers.

Each judge returned a short reason and one of `match`, `no_match` or `no_answer`. The grading protocol provides the input template and response schema.

B.3 Preregistration and Deviations

H1 and the adequacy threshold $\kappa_{\min} = 0.50$ were preregistered, but the cohort selection and bootstrap procedure were chosen during analysis. H2 and H3, background associations, classifier error limits and electricity scenarios are exploratory.

The study reference labels were estimated from 30 fresh answers per question rather than the preregistered 20. The realised question draw balanced category and five-generation success counts rather than additionally stratifying by list style. The preregistered judge-validation plan specified sampling by agreement score. The calibration sample described below was instead selected by category, response length and list style. The panel-agreement gate was added after preregistration. The reported power calculation assumes equal frequencies across the six success bands, rather than the empirical frequencies specified for the planned preregistered update.

B.4 Judge-Panel Calibration

Three authors and the three LLM judges independently graded 100 model responses. The sample combined 70 responses drawn across the 14 category categories with the 15 shortest responses, the 10 longest and five responses to list-style questions. This selection was intended to include potentially difficult grading cases, such as brief answers and long responses containing extensive reasoning.

Table 3: Question sets and their roles

Question set	Items	k	k_h	Main use
Retained pool	8,454	5	–	Screening labels and strata
Development	6,705	5	–	Classifier fitting and selection
Test set	1,655	5	–	Evaluation pool; format cross-evaluation
Study subset	280	30	3	Paired forecasts and scenarios
Test remainder	1,375	5	–	Classifier and scenario analyses

Responses were sampled individually, allowing several generations from the same question to be selected. The sample therefore covered 90 distinct questions. Most repeated questions occurred among the shortest responses, and some generations contained identical answer text.

For comparison with the authors, match counted as positive and the other two verdicts as negative. The panel’s majority agreed with the human majority on 90 responses, comprising 44 true positives and 46 true negatives. There were two false positives and eight false negatives. Scott’s $\pi = 0.79992$ was narrowly below the panel threshold of 0.80. Given the submission deadline and the time required to repeat grading, we proceeded with the study despite this shortfall. Individually, gpt-oss agreed with the human majority on 93 responses ($\pi = 0.8599$), while Ministral and Qwen each agreed on 90 ($\pi = 0.79992$). All three exceeded the individual threshold of 0.60.

The same 100 responses were graded again during the full grading run. The panel changed its majority verdict on two responses, reducing agreement with the human majority from 90% to 88% ($\pi = 0.7596$).

B.5 Conversion-Fidelity Audit

Preliminary converter selection. The process of selecting a suitable converter model employed an AI-assisted comparison of mistral-small-2603, deepseek-v4-pro and deepseek-v4-flash on 93 questions, supervised by one author. Across two runs, Flash changed six filtering decisions (6.5%), compared with 16 for Mistral (17.2%), supporting its selection as the converter.

Stage 1: assessing conversion decisions. A stratified sample of 248 original questions was assessed independently by all three authors. The sample included 96 questions accepted by the initial filter, 26 accepted after rescoring, 64 rejected after rescoring, 50 with borderline scores of 4–7, and 12 rejected during rewriting. The borderline group included both accepted and rejected questions around the readmission cutoff of 5 and was counted separately from the other groups.

Each question was assessed for whether it would retain a single correct answer without its options and whether the reference answer was suitable for grading. The original question, options and correct answer were shown, while the converter’s decision and rewrite remained hidden.

Stage 2: assessing the retained rewrites. The retained rewrites were then assessed for preservation of meaning and whether they could be understood without the answer options. This stage covered the 152 questions retained from the same sample, comprising the

96 initial acceptances, the 26 acceptances after rescoring and 30 retained questions from the borderline group.

Preservation of meaning. Eight rewrites were judged by majority decision to have changed the original meaning. These comprised 4 of the 96 initial acceptances, 2 of the 26 acceptances after rescoring and 2 of the 30 retained borderline questions.

Because the groups were sampled at different rates, their results were weighted to reflect their prevalence among all 8,454 retained questions. The corresponding group sizes were 7,704, 573 and 177. This gives an estimated meaning-change rate of 4.46%, with inter-rater agreement of Fleiss’ $\kappa = 0.572$. The estimate falls below the audit protocol’s 5% threshold.

Self-containedness. The yes/no labels for one annotator were corrected following confirmation that their meaning had been interpreted in reverse. After correction, 2 of the 96 initial acceptances, 1 of the 26 acceptances after rescoring and none of the 30 retained borderline questions were judged by majority decision to lack sufficient context to stand alone. Applying the same population weights gives an estimated rate of 2.16%. Inter-rater agreement remained low (Fleiss’ $\kappa = 0.121$). The correction did not affect the preservation-of-meaning assessment.

The separate judge-panel calibration is described in Appendix B.4.

B.6 Generation Settings

Both generation runs used the standard google/gemma-4-E2B-it checkpoint², with temperature 1.0, top- p 0.95, top- k 64, a 6,144-token output limit and an 8,192-token context limit. The default chat template was used without a preset system prompt. The standard checkpoint was chosen as a baseline to avoid introducing variation from a particular quantised version of the model. The measured success rates therefore describe this serving configuration and may differ from those obtained with quantised variants on consumer hardware.

C Human-Study Supplement

C.1 Participant Instructions

The instructions defined a satisfactory response as one that gives the known correct answer, regardless of wording, length or style. Examples showed that differently phrased numerical answers could both be correct, while a response that merely discusses the topic does not answer the question. Participants received this simplified explanation rather than the full grading rubric, which also specified a numerical tolerance and detailed decision rules.

²<https://huggingface.co/google/gemma-4-E2B-it>

Table 4: Participant characteristics. All 30 participants completed the forecast task.

Characteristic	Summary
Age, years (valid $n = 29$)	Mean 23.3 (SD 2.9); median 23 [IQR 21–25]; range 18–31
Not reported	1 (3.3% of cohort)
Gender (valid $n = 29$)	
Female	9 (31.0%)
Male	20 (69.0%)
Not reported	1 (3.3% of cohort)
Occupation/status (valid coded $n = 27$)	
Student (including employed)	25 (92.6%)
Employee only	2 (7.4%)
Nonblank, ambiguous	1 (3.3% of cohort)
Not reported	2 (6.7% of cohort)
Education, completed or current (descriptive; valid coded $n = 28$)	
Bachelor degree or current bachelor study	18 (64.3%)
Master	7 (25.0%)
High school	1 (3.6%)
Current state-examination study	1 (3.6%)
Completed master-equivalent state examination	1 (3.6%)
Nonblank, ambiguous	1 (3.3% of cohort)
Not reported	1 (3.3% of cohort)
AI-use frequency (valid $n = 28$)	
Less than weekly	1 (3.6%)
1–2 days/week	3 (10.7%)
3–4 days/week	8 (28.6%)
5–6 days/week	2 (7.1%)
Daily	14 (50.0%)
Not reported	2 (6.7% of cohort)

Participants were asked to judge how likely the local model was to answer each question satisfactorily, relying on their first impression rather than solving it themselves and they received no feedback on the model’s answers.

Participation was voluntary and required informed consent. Participants received 0.5 VP hours of course credit. The study and this paper were developed as part of the course “AI & Sustainability” at the University of Regensburg under the supervision of Prof. Dr. David Elsweiler, who also approved the study.

C.2 Sample and Allocation Details

The primary analysis includes one complete set of forecasts for each of the 30 assigned participant IDs. There were 34 submissions in total, including three additional complete sets and one incomplete set.

Forecasts were included even when background information was missing. The demographic summaries in Table 4 therefore use the available responses for each field, so their sample sizes can differ.

Questions were selected to balance subjects and success bands, but some combinations had too few eligible questions. Of the 84 subject–band combinations, 12 contained fewer questions than

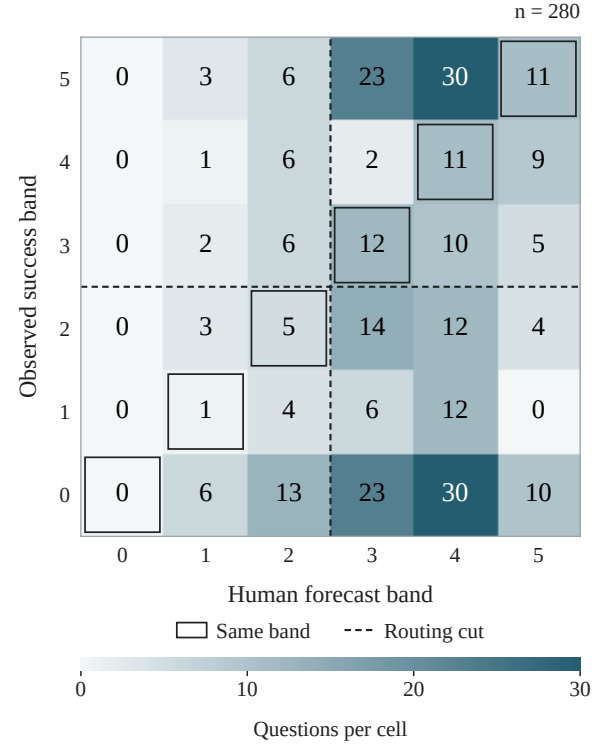


Figure 1: Human forecasts versus reference success on 280 questions. Human bands use the median of three forecasts. References use 30 local-model answers per question.

planned. Additional questions were selected from nine other combinations to complete the sample.

C.3 Forecast Agreement and Robustness Checks

Weighted agreement measures how closely human forecasts match the reference success bands. Kendall’s $\tau_b = 0.08$ additionally describes whether questions given higher human forecasts also fall in higher reference bands. Ordinal Krippendorff’s α measures agreement among participants [26]. Bootstrap intervals were calculated by resampling questions while keeping the participant sample fixed. They therefore describe uncertainty across questions, rather than across different samples of participants.

Two further analyses checked whether individual participants had a large influence on the results. First, the analysis was repeated with each participant left out in turn. Second, participants with unusually high or low individual κ_w scores were excluded. Two rules were applied separately to identify these participants. The interquartile rule flags scores outside $[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$. The robust-SD rule flags scores more than two robust standard deviations from the median, where robust SD is 1.4826 times the median absolute deviation.

Exclusions can leave a question with two different ratings and therefore no single middle rating. In these cases, agreement was calculated twice, using the lower remaining rating and then the

upper. Questions were retained whenever at least one forecast remained.

C.4 Reference-Label Stability and Sensitivity Analyses

Changes in reference labels. The estimates based on five generations and those based on 30 fresh generations assigned 49.3% of the 280 study questions to the same success band. More questions fell into the extreme bands in the later run: the number in the lowest band increased from 56 to 82, and the number in the highest from 39 to 73. The binary local/cloud labels remained unchanged for 86.4% of questions.

Uncertainty from repeated generation. A success rate estimated from a finite number of answers is subject to sampling variation. Assuming independent answers with a fixed probability of success, the maximum standard error of $\hat{p}$ is $\sqrt{0.25/k}$, where k is the number of generations. Increasing k from 5 to 30 reduces this maximum from 0.224 to 0.091.

A simulation examined how this sampling variation affects agreement with the reference labels. A binomial mixture was fitted to the five-generation success counts. Each simulated sample contained 280 questions, balanced across those initial success counts, with 30 new generation outcomes simulated per question. A simulated forecaster that knew each question’s underlying success probability achieved mean weighted agreement of $\kappa_w = 0.899$ with the simulated reference labels across 6,000 repetitions.

Sensitivity to individual raters. When each participant was left out in turn, agreement with the reference labels ranged from $\kappa_w = 0.025$ to 0.086. Excluding the five raters flagged by the robust-SD rule described above gave $\kappa_w = 0.048$ when the lower of two remaining ratings was used and 0.034 when the upper was used.

C.5 Exploratory Predictors of Forecast Quality

We also explored whether differences in forecast agreement or overestimation were associated with participants’ backgrounds. The analyses considered age, completed qualification, student status, GenAI use and gender. Each was examined separately using 25–29 available responses. A model combining age, qualification, student status and GenAI use included 23 participants with complete responses.

Neither approach gave a clear result. In the separate models, the confidence intervals spanned zero and included associations in both directions; the combined model was also inconclusive. The samples were small and imbalanced, so these findings do not show that participant background is unrelated to forecast quality. The estimates, variable coding and details of the HC3 confidence intervals [36] are available in the supporting results.

D Classifier Data, Training and Comparisons

D.1 Corpus and Response Distributions

The local model was evaluated in both multiple-choice and open-ended form on the same 8,454 retained questions, with five generations per question. In multiple-choice form, it answered 48.2% of questions correctly in all five generations and 18.5% in none,

averaging 3.28 correct answers out of five. In open-ended form, the corresponding proportions were 33.4% and 38.7%, with an average of 2.36 correct answers out of five.

In both formats, most questions were therefore answered correctly either in all five generations or in none: 66.7% in multiple-choice form and 72.1% in open-ended form.

Benchmark exposure. The local model’s reported training cutoff falls after the release of MMLU-Pro [23, 49]. Some measured success may therefore reflect prior exposure to benchmark questions [34].

D.2 Dataset Splits and Exclusions

To prevent questions with identical original text from appearing in different splits, we excluded 91 of the 8,454 converted questions. Where such overlaps occurred, test items took priority over validation items, and validation over training. This left 5,851 training, 857 validation and 1,655 test questions.

Three further training questions were excluded because their converted text duplicated questions in validation or test. Combining the remaining training and validation questions gave 6,705 questions for classifier development. Questions linked by identical original or converted text were also grouped into families, and no family containing a test question was allowed in development.

D.3 Comparing and Selecting Prediction Heads

The prediction head turns the classifier’s representation of a question into a success forecast. We compared seven heads trained to predict five-generation success counts. Binomial and beta-binomial heads predict probabilities for the possible counts, with the beta-binomial head also learning a concentration parameter. CORN treats the count as an ordered outcome [44]. Two softmax heads predict a distribution over the six possible counts, using either cross-entropy or a normalised ranked probability score as the training loss [22]. The remaining two heads use direct regression and a joint softmax-binomial prediction.

The comparison proceeded in five stages and comprised 108 fits. A pilot using three seeds eliminated four heads. For the remaining three, we screened a limited set of combinations of learning rate, weight decay and LoRA dropout. The best settings were then replicated across seeds and compared using five-fold cross-validation, keeping questions from the same text family together. Each fold used an inner validation split to select the checkpoint evaluated on the held-out fold.

Table 5 reports mean per-generation log loss for the six configurations evaluated in cross-validation. Scores were calculated by pooling out-of-fold predictions for the 6,705 development questions. The highlighted beta-binomial configuration had the lowest loss and was selected on that basis. The best configuration for each of the three heads was then refit on all development data across three seeds, using the epoch chosen through cross-validation.

The selected beta-binomial classifier was refit for eight epochs with a 20-epoch learning-rate schedule. Its adapter and tokenizer are available alongside the complete recorded recipe.

D.4 Classifier Evaluation

This evaluation uses the 1,375 questions remaining after excluding the 280 human-study questions from the 1,655-question test split.

Table 5: Cross-validation results for two configurations of each finalist head. The selected beta-binomial configuration had the lowest loss and is shown in bold. Learning rates are rounded.

Head	Learning rate	Weight decay	LoRA dropout	Log loss
Beta-binomial	9.47×10^{-5}	0.10	0.10	0.58867
Beta-binomial	5.00×10^{-5}	0.01	0.10	0.59356
Binomial	1.54×10^{-4}	0	0.05	0.59068
Binomial	5.00×10^{-5}	0.01	0.10	0.59225
Softmax (RPS)	1.15×10^{-4}	0.01	0.05	0.59160
Softmax (RPS)	5.00×10^{-5}	0.01	0.10	0.59165

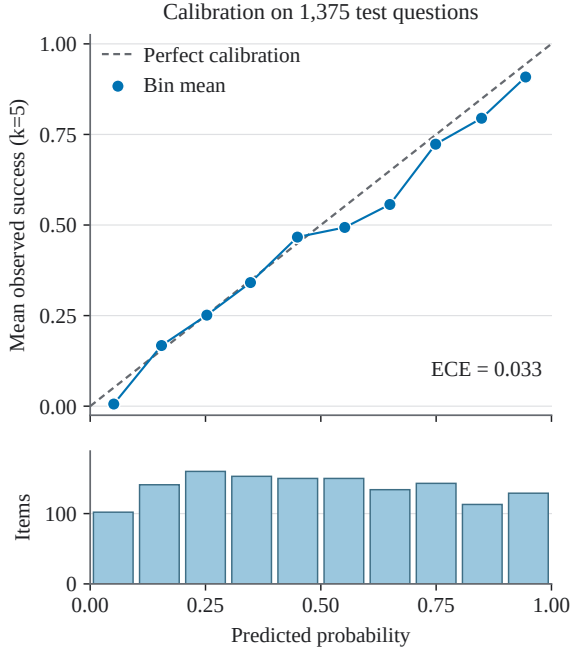


Figure 2: Predicted versus observed local-model success on 1,375 test questions. Points average predictions and five-generation success fractions within ten equal-width probability bins. Bars below show the number of questions per bin. ECE is the bin-weighted mean absolute gap between predicted and observed success.

Predicted success probabilities were compared with each question’s success fraction across five generations, rather than with binary routing labels. Mean per-generation log loss was 0.539, probability RMSE was 0.377, and expected calibration error was 0.033. Calibration was assessed using ten equal-width probability bins (Figure 2). Routing performance was also evaluated across distinct prediction thresholds and at the all-local and all-cloud extremes.

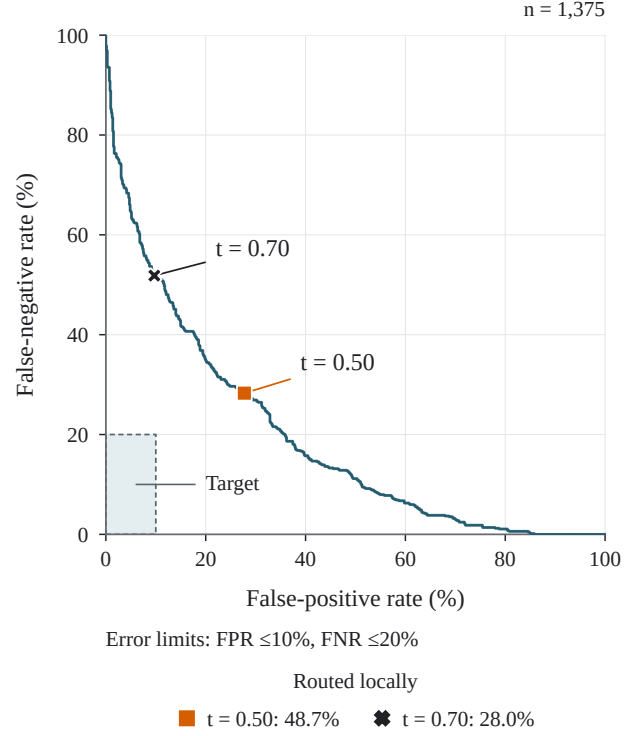


Figure 3: Classifier routing errors on 1,375 test questions with five-generation reference labels. None of the possible thresholds meets both error limits. False-positive and false-negative rates use 721 reference-negative and 654 reference-positive questions, respectively.

D.5 Comparing Routers Across Question Formats

We also compared the multiple-choice-trained router and the selected open-ended-trained router on questions with and without answer options. Both were evaluated on the same 1,655 test questions, using each format’s own five-generation success counts (Table 6). In both formats, the router trained on that format had higher agreement and routing accuracy, and lower log loss.

The two routers differ in more than their training question format. Both use ModernBERT and a beta-binomial likelihood, but differ in head parameterisation, training schedules and checkpoint selection. It must also be noted, that open-ended-trained router was refit on the combined development set, whereas the multiple-choice-trained router used a separate validation set to select a checkpoint. The comparison therefore cannot fully isolate the effect of question format. This comparison therefore serves as an exploratory one.

D.6 Comparison with Prior Routing Approaches

The approaches discussed in Section 2.1 differ in both when they route a request and what their scores predict (Table 7). Decisions

Table 6: Cross-format performance on 1,655 test questions, using each format’s five-generation labels. MC denotes multiple-choice; routing accuracy uses threshold 1/2.

Training format	Test format	κ_w	Accuracy	Log loss
Open-ended	Open-ended	.366	70.8%	.568
MC	Open-ended	.127	56.7%	.691
Open-ended	MC	.183	57.9%	.712
MC	MC	.248	72.4%	.567

made after generation can use the answer itself; decisions made beforehand cannot.

Table 7: Routing approaches by decision timing, available information and prediction target.

Approach	Decision	Signal	Score predicts
FrugalGPT [12]	after gen.	query + answer	correctness
RouteLLM [39]	before gen.	query	which model wins probability of meeting a relative-quality criterion
Hybrid LLM [15]	before gen.	query	
CR ² device gate [51]	before gen.	query emb. cost weight	utility margin reference-match probability
This work	before gen.	query	

E Electricity Scenarios

E.1 Routing and Answer Success

Let $a_i = 1$ select the local model for question i and $a_i = 0$ select the cloud. Human routing follows the majority of three ratings, while classifier routing uses a predicted-success threshold of 1/2. The observed fraction $s_i = c_i/k$ supplies local answer success, separately from that routing forecast. We use $k = 30$ for the paired study questions and $k = 5$ for the test remainder. Cloud success is fixed at one.

Let L_i and C_i be electricity per local and cloud attempt, and H the classifier overhead per question, set to zero for human and all-cloud routing. Accepting the selected model’s first answer gives

$$\bar{E}_A = \frac{1}{N} \sum_i [H + a_i L_i + (1 - a_i) C_i], \quad (1)$$

$$Q_A = \frac{1}{N} \sum_i [a_i s_i + (1 - a_i)]. \quad (2)$$

Here $\bar{E}$ is expected Wh per question, while Q is expected answer success. Always-cloud inference has $\bar{E}_C = N^{-1} \sum_i C_i$ and $Q_C = 1$. Electricity savings are $\bar{E}_C - \bar{E}_A$, or $1 - \bar{E}_A / \bar{E}_C$ as a fraction. Averaging across questions weights each category by its number of questions.

E.2 Recognised Failures and Retries

If a failed local answer can be recognised, it may instead be retried locally before falling back to the cloud. With at most $n = 1, \dots, 6$ local attempts (zero to five extra retries), the expected number of

attempts is

$$A(s_i, n) = \sum_{j=0}^{n-1} (1 - s_i)^j. \quad (3)$$

The cloud is used only if every permitted local attempt fails. Hence

$$\bar{E}_B = \frac{1}{N} \sum_i [H + (1 - a_i) C_i + a_i \{L_i A(s_i, n) + (1 - s_i)^n C_i\}]. \quad (4)$$

Success is $Q_B = 1$ because any unresolved question reaches the perfect cloud reference. Both direct and fallback cloud calls therefore succeed in one attempt at cost C_i . This is an optimistic recovery scenario, since the study did not evaluate failure recognition.

The simulation samples local attempts independently at each fixed s_i , stopping at the first success or the attempt limit. We use 10,000 repetitions and check the means against the exact expectations above. Simulation standard errors describe numerical precision, not uncertainty in the success rates estimated from five or thirty answers.

E.3 Electricity References

Costs were estimated with EcoLogits 0.11.1 using its model catalogue, output-token counts and generation-latency estimates [16, 17]. The cloud reference is gpt-5.5-pro-2026-04-23, with gpt-5.5-2026-04-23 providing a lower-cost comparison. The local reference is gemma-4-26b-a4b-it and it serves as a referenced cost proxy. The estimates include GPU and allocated non-GPU server electricity, with a PUE adjustment for datacentre cooling. They exclude networking, storage and provider services such as backups and monitoring. Full input-processing and end-user-device costs are not modelled.

For each question, the token count is its mean observed local output length. Cloud length is also evaluated at 0.5 and 2 times this mean. These lengths are scenario inputs because cloud answers were not collected. The same cost is used for each local retry of a question. Savings compare corresponding lower or upper EcoLogits estimates in both policies, rather than subtracting unrelated endpoints. Variation across output lengths or cloud models is reported separately from those estimator ranges.

Classifier overhead was measured over three sequential passes of all 1,655 test questions on an Intel Core Ultra 7 258V laptop, using CPU inference in FP32 with four threads and batch size one. RAPL package and DRAM counters recorded each pass and a subsequent idle period of equal duration, with the model loaded. The laptop remained on battery in balanced mode, with unrelated applications closed, supposed to mimic a natural working system usage environment. Subtracting idle use from the mean gross electricity of 0.000387 Wh per question gave $H = 0.000221$ Wh, with individual passes spanning 0.000218–0.000224 Wh. These counters cover the processor package and DRAM, not whole-device wall electricity.

E.4 Category Results and Scenario Comparisons

Table 8 reports classifier routing at threshold 1/2 for the 1,375 test questions outside the human study, broken down by category. Expected success is calculated using individual questions’ success rates before averaging within each category. Category sizes reflect the questions remaining after selection for the human study. These benchmark categories do not represent patterns of everyday

use, and results for small categories should not be read as precise estimates for their fields.

Figure 4 compares accepting the selected model's first answer with sending every recognised local failure directly to the cloud. Under the assumptions of perfect failure recognition and cloud success, fallback gives 100% expected answer success but uses more electricity. Allowing additional local attempts before cloud fallback reduces expected electricity use in these scenarios (Figure 5).

The choice of cloud reference substantially changes the absolute electricity estimates. On the test remainder, using standard GPT-5.5 instead of Pro gives 1.93–4.00 Wh per question when accepting the selected model's first answer and 2.52–5.22 Wh when recognised local failures are sent directly to the cloud.

Across both cloud references and all three output-length settings, accepting the first answer saves 44.0% to 45.6% relative to all-cloud inference on the test remainder. With recognised failures and zero to five extra local retries before cloud fallback, savings range from 27.2% to 33.4%. These percentage ranges summarise variation across scenarios; the lower and upper EcoLogits estimates describe the cost range within each scenario.

Table 8: Classifier routing by subject on the 1,375-question test remainder. Success and savings refer to accepting the selected model’s first answer; savings are relative to all-cloud inference. Fallback assumes every failed local answer is recognised and sent directly to a perfect cloud, giving 100% expected success. Ranges reflect EcoLogits cost bounds, using GPT-5.5 Pro with equal local and cloud output lengths. Overall values are averaged across questions.

Subject	N	Sent local (%)	Expected success (%)	All-cloud Wh/question	Selected answer Wh/question	Saved (%)	With fallback Wh/question
Overall	1375	48.7	84.7	108.17–288.74	59.14–158.38	45.1–45.3	77.23–206.92
Biology	58	58.6	72.1	85.42–224.48	32.35–84.39	62.1–62.4	55.96–146.38
Business	110	65.5	77.1	72.70–188.54	27.23–71.07	62.3–62.5	46.27–121.01
Chemistry	166	45.8	79.4	139.42–377.01	85.66–232.81	38.2–38.6	112.34–304.71
Computer science	28	89.3	87.9	92.14–243.44	11.32–30.10	87.6–87.7	21.62–57.15
Economics	72	73.6	75.8	99.98–265.59	26.06–69.10	73.9–74.0	52.80–140.59
Engineering	150	25.3	93.1	187.33–512.32	153.69–421.53	17.7–18.0	164.74–451.57
Health	50	28.0	90.8	61.02–155.55	44.36–113.20	27.2–27.3	49.30–125.60
History	5	0.0	100.0	44.43–108.70	44.43–108.70	0.0	44.43–108.70
Law	147	2.7	98.4	102.03–271.40	98.88–262.94	3.1	100.63–267.61
Math	207	92.8	85.9	110.20–294.47	8.56–22.87	92.2	33.69–91.49
Other	79	32.9	85.3	48.63–120.55	25.85–61.74	46.8–48.8	36.38–89.01
Philosophy	47	10.6	97.9	52.11–130.39	45.91–114.65	11.9–12.1	47.51–118.82
Physics	200	57.5	76.6	114.78–307.42	55.72–150.20	51.1–51.5	84.16–226.61
Psychology	56	26.8	86.4	79.20–206.90	51.60–133.42	34.8–35.5	66.07–172.02

Electricity and answer success

GPT-5.5 Pro costs with equal local and cloud output lengths

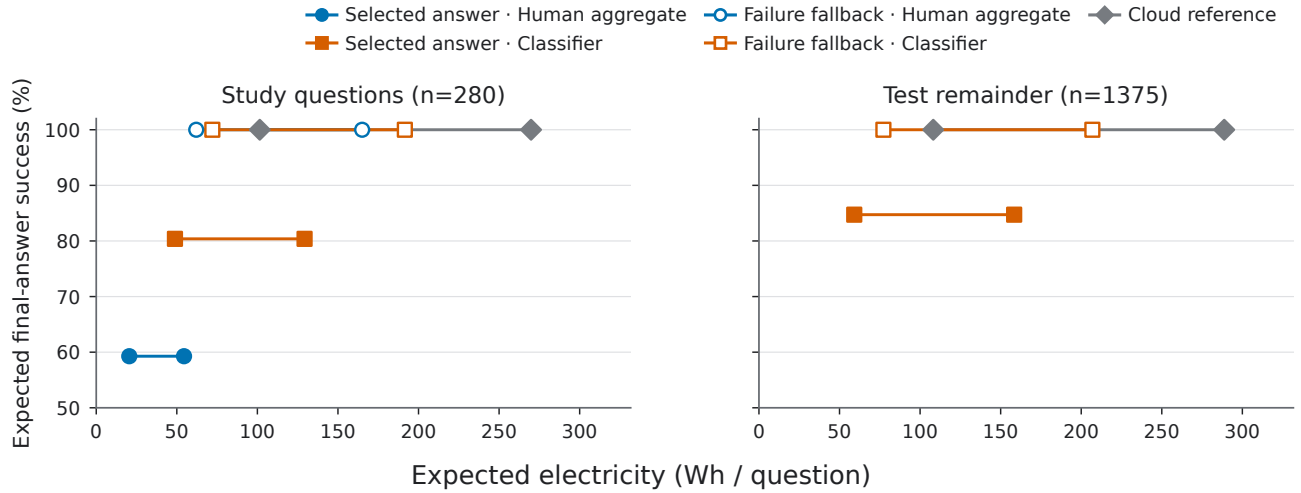


Figure 4: Expected electricity use and answer success on the study questions (left) and test remainder (right). Filled markers show results when the selected model’s first answer is accepted. Hollow markers assume every failed local answer is recognised and sent to a perfect cloud. Horizontal spans show the lower and upper EcoLogits estimates for GPT-5.5 Pro with equal local and cloud output lengths.

Electricity with local retries

GPT-5.5 Pro costs with equal local and cloud output lengths

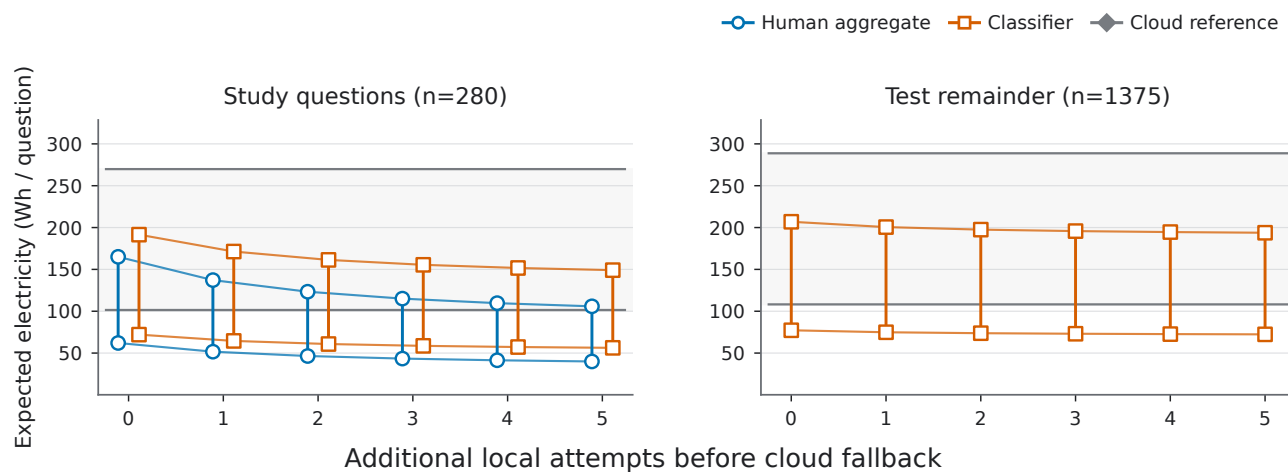


Figure 5: Expected electricity use as more local retries are allowed before cloud fallback. Retries stop at the first successful answer; unresolved questions go to the cloud once the retry limit is reached. Perfect failure recognition and cloud success are assumed, giving 100% expected success throughout. Vertical spans show the lower and upper EcoLogits estimates for GPT-5.5 Pro with equal local and cloud output lengths.

F Data and Code Availability

The accompanying research GitHub repository³ contains the grading, analysis and figure-generation code, along with detailed results and instructions. The LocalGate extension and daemon will be made available in the application repository.⁴ The datasets and the trained classifier are hosted under the LocalGate organisation on HuggingFace.⁵ We decided to not make the participant-level responses publicly available as to protect participants’ privacy. Anonymised responses may be requested from the authors. The study interface participants faced, is available in a separate repository.⁶

G Use of AI Tools

Large language models assisted with revising prose and structure, $\LaTeX$ formatting, and developing and debugging code. Model-suggested references were checked against primary sources, and quantitative claims about prior work were checked against the original publications. All manuscript text was reviewed by at least one author, and the authors take responsibility for the paper’s content and conclusions. The research questions were developed without AI assistance. Model use for question conversion, answer generation and grading is described in the methods and evaluation section.

³<https://github.com/LocalgateOrg/localgate-research>

⁴<https://github.com/LocalgateOrg/localgate>

⁵<https://huggingface.co/localgate>

⁶https://github.com/LocalgateOrg/Study_LocalModel